



INTRODUCTORY PAPER



2019 MATHEMATICS

**DO NOT OPEN THIS BOOKLET
UNTIL INSTRUCTED.**

30 QUESTIONS

TIME ALLOWED: 35 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a pencil. Do **NOT** use a coloured pencil or a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

There are **30 MULTIPLE-CHOICE QUESTIONS** (1–30).

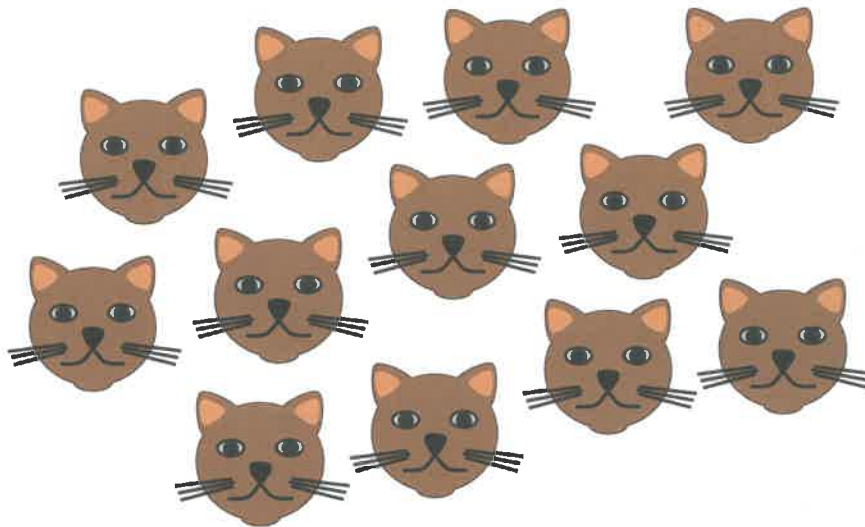
Use the information provided to choose the **BEST** answer from the four possible options.

On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a ruler and spare paper.

You are **NOT** allowed to use a calculator.

1. How many cats are in this picture?



- (A) 10 (B) 11 (C) 12 (D) 13

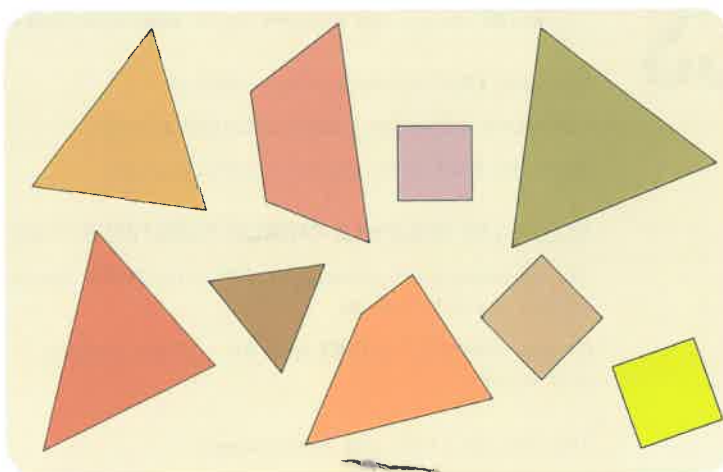
2. Oscar is counting up by the same number each time.

2, 8, 14, 20, 26, 32, ?

What number must ? be?

- (A) 33
(B) 34
(C) 36
(D) 38

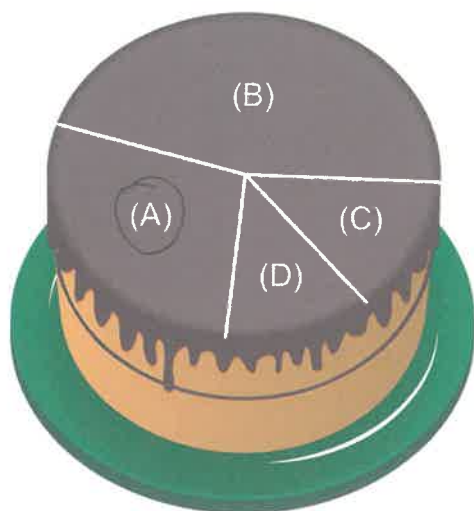
3. How many triangles are in this picture?



- (A) 4 (B) 5 (C) 7 (D) 8

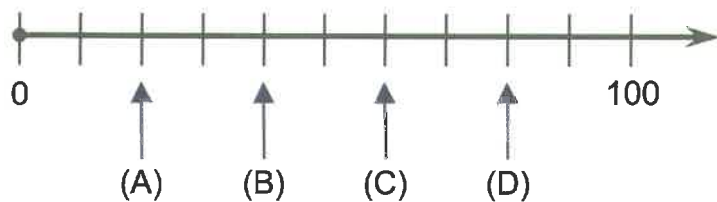
4. Amy took the largest piece of cake.

Jack took the next largest piece.



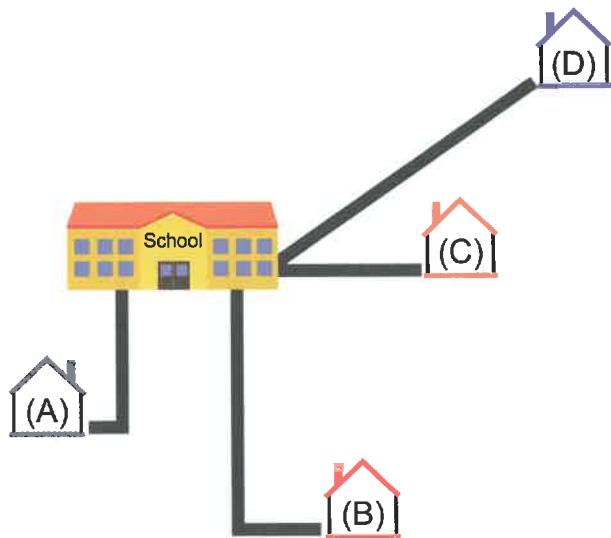
Which piece of cake did Jack take?

5. Which arrow is pointing to 40 on this number line?



6. The map shows the houses of four children near a school. Each child walks to school along a path.

Which house has the shortest path to school?



7. Jake's birthday is on Friday.

What is the day after his birthday?

(A) Monday (B) Wednesday (C) Thursday (D) Saturday

8. This chart shows the shells some children collected at the beach.

Name	Shells
Peter	
Heidi	
Sue	
Henry	

How many shells did Sue collect altogether?

(A) 4 (B) 5 (C) 6 (D) 8

9.

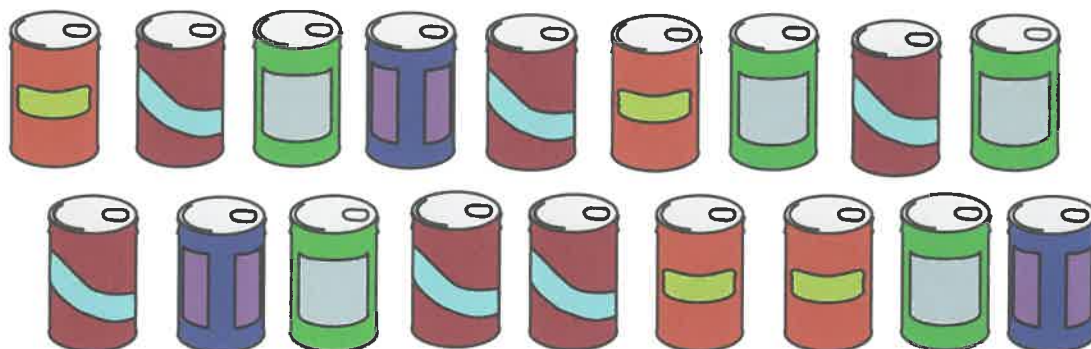
$$\text{Blue ring} = 5 \quad \text{Red ring} = 7$$

$$\text{Blue ring} + \text{Red ring} + \text{Blue ring} + \text{Red ring} + \text{Blue ring} = \boxed{?}$$

What value must $\boxed{?}$ be?

- (A) 12 (B) 14 (C) 15 (D) 29

10. Ella picks one of these cans without looking.



Which kind of can is she most likely to pick?



(A)



(B)

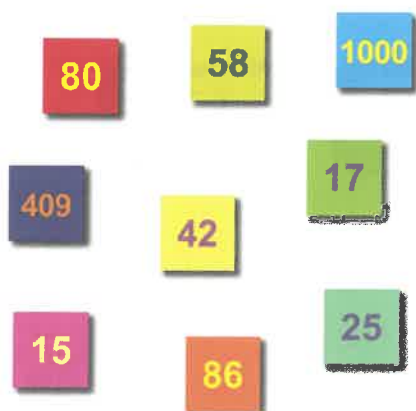


(C)



(D)

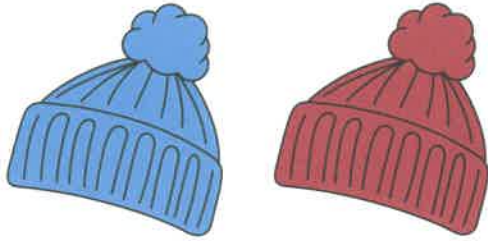
11. Chandra has these cards.



How many cards have a number greater than 80 written on them?

- (A) 2 (B) 3 (C) 4 (D) 6

12. Sue has two hats.



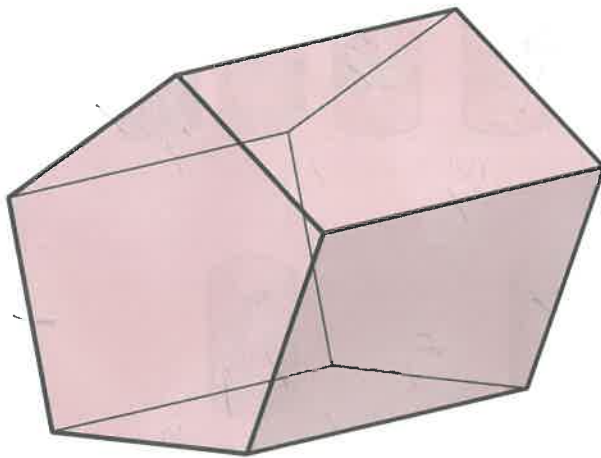
She has twice as many scarves as she has hats.

She has twice as many shirts as she has scarves.

How many shirts does she have?

- (A) 8 (B) 6 (C) 4 (D) 2

13. How many edges does this object have?



- (A) 5 (B) 7 (C) 10 (D) 15

14. These clocks show a pattern of times.



Which of these must replace  to fit the pattern?

- (A)  (B)  (C)  (D) 

15. Toby has 21 apples. Lillian has 4 apples fewer than Toby.

How many apples does Lillian have?

- (A) 4 (B) 17 (C) 18 (D) 25

16. The table shows some information about the books some children read.

	Non-fiction	Fiction
Boys	20	16
Girls	5	15

How many children read fiction books?

- (A) 15 (B) 16 (C) 31 (D) 56

17. $120 - 11 = ?$

- (A) 131 (B) 119 (C) 111 (D) 109

18. Some students were walking along a track.

The table shows how far they had walked after 10 minutes.

Student	Distance
Sally	605 metres
Joe	575 metres
Nadia	520 metres
Billy	580 metres
Ellen	610 metres
Alex	565 metres

Who walked 15 metres further than Alex?

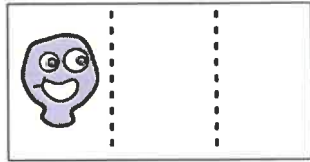
- (A) Joe (B) Nadia (C) Billy (D) Ellen

19. Sue is 180 cm tall. Mel is half as tall as Sue.

How tall is Mel?

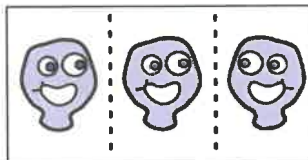
- (A) 40 cm (B) 90 cm (C) 100 cm (D) 140 cm

20. Sarah used this face to make a pattern. She flipped it over the dotted lines.

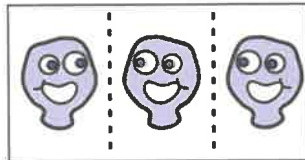


Which picture shows the result?

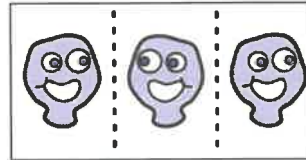
(A)



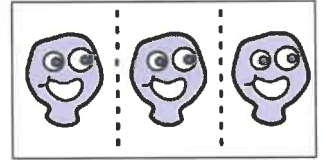
(B)



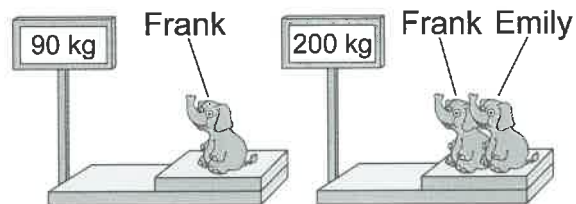
(C)



(D)



21. Emily and Frank are two baby elephants.



What is Emily's mass?

- (A) 90 kilograms (B) 110 kilograms (C) 190 kilograms (D) 200 kilograms

22. In Mr Zu's class you get a silver star for doing good work.
If you have five silver stars, you get a gold star.

Gold Stars



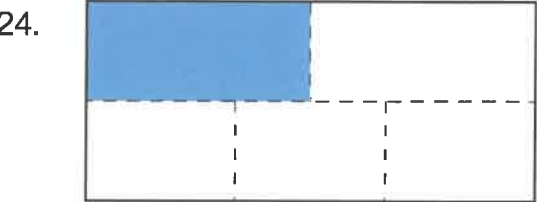
Who has 15 silver stars?

- (A) Sue (B) Mike (C) Sam (D) Debbie

23. ? + 15 = 28 - 8

What is the missing number?

- (A) 5 (B) 10 (C) 20 (D) 35



What fraction of this shape is shaded?

- (A) $\frac{1}{2}$ (B) $\frac{1}{4}$
(C) $\frac{1}{5}$ (D) $\frac{1}{3}$

25. Each month Anna paints 10 pictures and sells them all.
Each month Tim paints 16 pictures and sells half of them.
In three months, how many paintings do Anna and Tim sell altogether?
- (A) 26 (B) 38 (C) 54 (D) 78

26. Sasi poured a full jug of water into each of these containers.
Which container can hold more than the jug?

(A)



(B)



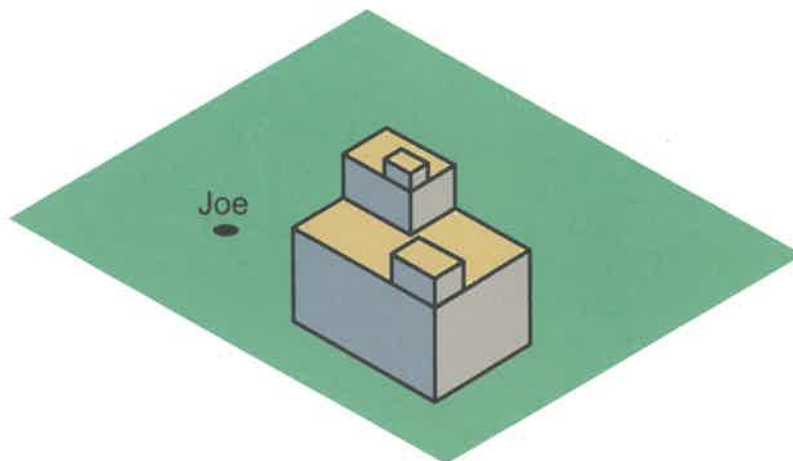
(C)



(D)

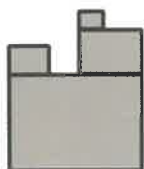


27. Joe is standing near a building as shown.

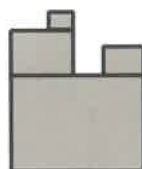


Which of these best shows Joe's view of the building?

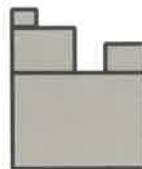
(A)



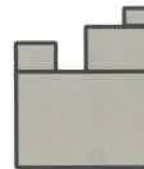
(B)



(C)



(D)



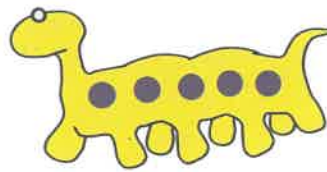
28. Jane is playing a computer game. In the game there are monsters of different sizes.



1 spot
4 legs



3 spots
6 legs



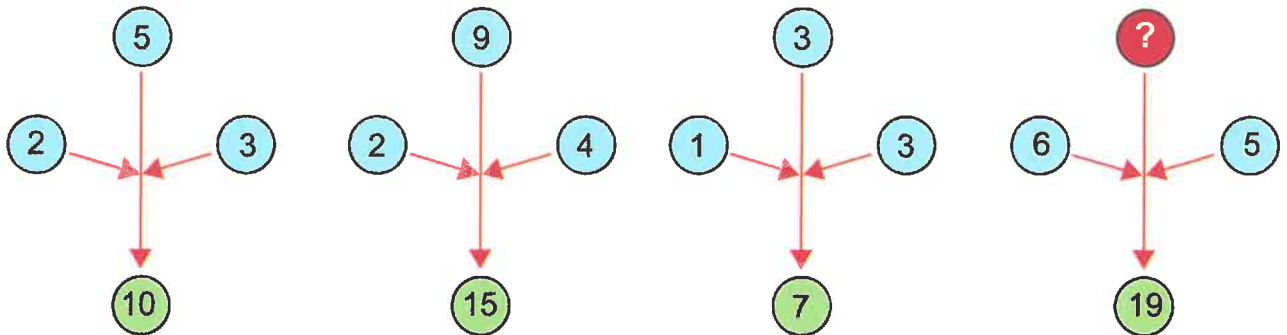
5 spots
8 legs

Bigger monsters follow the same pattern.

How many spots would a monster with 16 legs have?

- (A) 10 (B) 11 (C) 13 (D) 19

29.



What must the value of ? be?

- (A) 8 (B) 11 (C) 28 (D) 30

30. Eliza picked two red roses and five yellow roses and put them on the table.



She closed her eyes and mixed them up.

What is the smallest number of roses Eliza must pick up to be certain she picks up a red one?

- (A) 2 (B) 3 (C) 5 (D) 6